PA-consistency machines: comparison and new reduction experiments

Research report prepared for Vladimir Reshetnikov

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Abstract

The requested endpoint is a binary, single-tape Turing machine with at most 195 working states, started on a blank tape, whose nonhalting is provably equivalent in PA to consistency of ordinary first-order PA. **This investigation did not obtain a verified machine meeting that endpoint.** The smallest established reference table remains the published 196-state construction. This report and its archive are a research result, not a claim to have constructed a 195-state machine.

We recheck the supplied 208-state collection, the specified 209-state predecessor, and PA196 against frozen source identities. The complete accepted U208 collection has twenty rooted graph-isomorphism classes; its final deletion is not an additional deletion available in PA196. New work includes checked combinations of source rewrites, stronger physical deletion searches, an equivalent split of the first quantifier head with explicit Hilbert certificates, and two full-schema countermodels to tempting axiom deletions. We also identify a precise HA-strength modification of the supplied iPRA core and distinguish the published one-way internal PA theorem from the requested biconditional. All computational evidence is classified by its actual verification scope. No unsuccessful finite search is represented as a lower bound on all smaller machines.

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1 The target and the evidence boundary

The intended theorem is

$$\mathrm{PA} \vdash (\neg\,\mathrm{Halt}(M) \leftrightarrow \mathrm{Con}(\mathrm{PA})), \qquad |Q_M| \leq 195.$$

Here PA has full first-order induction. The machine has alphabet $\{0, 1\}$, a two-way infinite tape initially filled with zeros, one transition for each working state and read symbol, and one-cell left or right moves. The distinguished halting control is excluded from $|Q_M|$. These are the common conventions of the supplied constructions. A larger alphabet, a nonblank initial program, or a change in how halt is counted would require a different comparison.

Input	Frozen revision
ProveIt	`dc523e7b06ffb414093c90aa035500ca283e5f75`
BusyBeaver, PA196	`407cdc356f6b586635e71ac92be394255f4bfea4`
BusyBeaver, requested PA209	`69d98241a721ead20b4f0b17df9062950b40fbfe`

The current ProveIt incoming directory includes earlier responses to this same research problem. They were inspected before the new searches, and their results are attributed as prior work. In particular, an alternative 196-state decoder, the shorter-negation experiment, distributed separation, diagonal vacuity, and the MP-success rewrite were already present. Replaying them does not make them new discoveries. The archive preserves the relevant original license and provenance notices. No source repository was modified or published.

The report uses four distinct kinds of evidence:

1. Exact finite checks: table sizes, graph isomorphisms, input hashes, complete finite startup traces, and finite proof certificates.

2. Symbolic instruction checks: a specified executor checks arbitrary register values under its explicit engine contracts. This is stronger than testing a few numerical inputs, but the executor and its binding still belong to the trusted argument.

3. Mathematical proofs: the split-head equivalence and the countermodels below are schema-level arguments. Their accompanying finite controls supplement those arguments.

4. Unfinished endpoints: no new end-to-end Lean or Rocq theorem of the requested PA biconditional, and no accepted sub-196 table, is delivered here.

2 Comparison of the supplied constructions

The comparison was performed against ProveIt revision `dc523e7b06ffb414093c90aa035500ca283e5f75`, the requested PA209 revision `69d98241a721ead20b4f0b17df9062950b40fbfe`, and the published PA196 table [1, 3, 4]. The latter's SHA-256 is `e8bbccd4cfa325251ca95066db116b5aa766e404b0971c33c02772be1c3617a8`. State counts throughout count working controls; the distinguished halt control is separate. Each machine is binary, has one two-way infinite tape, starts from all zeroes, and moves one cell left or right on every transition.

The classification and the earlier reductions described in this section already occur in the incoming research. They have been freshly replayed and their input bytes rebound to the current original ZIP members; they are not new state reductions discovered in this investigation.

2.1 The full U208 collection has twenty graph classes

The nine headline tables in the five original U208 packages form exactly two rooted transition-graph isomorphism classes, called A and B. Each isomorphism preserves the initial and halting controls, read symbols, written symbols, and movement directions. Raw byte hashes differ under renumbering, so comparing hashes alone would overcount independent designs.

Package, dated 29 October 2026	Principal table	Alternative
`U208_ConPA`	A	—
`U208_ConPA_Conditional_Context`	A	B
`U208_ConPA_Entry_Context`	A	B
`U208_ConPA_History`	A	B
`u208_research`	B	A

The complete collection is larger. The archive `pa208_research_2026-10-29.zip` contains eighteen additional accepted 208-state helper constructions, nine based on each class. All eighteen graphs are pairwise nonisomorphic and differ from A and B. Consequently the accepted U208 material audited here consists of *27 presentations in 20 rooted graph classes*. Each new helper construction deletes one old state but introduces one fresh helper, so its net state count stays 208.

The retained comparison checks 30 selected literal inputs: these 27 U208 tables, Chang's PA209 and PA196, and the previously supplied alternative196. It reconstructs all sixteen isomorphism maps among the nine headline tables and verifies every transition under each map. It also independently rechecks the exact patches, incoming-edge predicates, and positive local whole-tape simulations for all eighteen helper variants. All 30 table hashes were checked directly against their current original archive members or repository files, rather than only against copied inputs. The portable results are in `comparison/verified_comparison.json`.

2.2 Source sizes and physical sizes

Design	States	Sites	Internal nodes	Leaf types	DAG total	Max. path
U208 A	208	3909	174	23	197	42
U208 B	208	3909	174	23	197	41
Chang PA209	209	3948	173	23	196	33
Chang PA196	196	3812	159	23	182	38
Incoming alternative196	196	3812	159	23	182	36

A source site is a primitive instruction occurrence. Internal nodes count distinct binary subtree pairs, while the total DAG also counts distinct primitive instructions and their operands. Thus U208's total DAG count 197 must not be compared directly with PA196's internal-node count 159.

A and B have the same ordered primitive word and the same exact ordinal control-flow graph; 720 binary instruction addresses differ. Chang's PA209 is distinct from the U209 parents inside the U208 reports. Sharing an earlier tape engine does not identify their literal state labels or make their source optimizations additive.

The PA196 budget is

$$36 \text{ engine states} + 13 \text{ leaf routers} + 147 \text{ source readers} = 196.$$

The 147 readers implement 159 internal source nodes. In particular, the six contextual identifications in the axiom body are already included in 196; they implement 68 axiom-grammar nodes using 62 physical readers.

The replay also checks the complete finite initializations, not merely a tape prefix. PA209 reaches its clean frame after 58,210 transitions; PA196 does so after 58,154. Their frames agree up to the

documented control-label correspondence: the head is at position -2, the register file represents candidate 2, the gap is 48, and there are 26 spare blocks of value 1. These are initialization facts, not evidence of nonhalting.

2.3 Why the final U208 deletion does not directly transfer

In the Joint native U209 numbering, the final U208 reduction identifies roles 172 and 211. An entry to 172 has predecessor bits 10; when 211 scans zero, its immediately preceding bit is one. Re-reading that history lets the merged control select the correct continuation. The replacement blocks for old state/read pairs $(172, 0), (172, 1), (211, 0), (211, 1)$ have lengths $7, 1, 3, 1$. Their endpoints agree on the entire tape and head position, not just on a finite numerical register sample. Thus the exact simulation clock is

$$T_{208} = T_{209} + 6N_{172,0} + 2N_{211,0},$$

where the N's count visits in the parent execution. The incoming-edge predicates and positive block lengths support a PA-formalizable finite stuttering argument, independently of the underlying theory calibration.

PA209-to-PA196 changes the axiom emitter and its coding. It transposes the arguments of the universal-quantifier Cantor pair, removes an old axiom head, adds guarded equality vacuity, and changes formula and emitter ordering. In the native comparison, 75 old controls disappear and 62 new ones are introduced: $209 - 75 + 62 = 196$. Roles 172 and 211 are already absent. Moreover, a complete finite scan of PA196 finds no state p whose two rows have the form

$$(p, 0) \mapsto (0, L, q), \qquad (p, 1) \mapsto (1, R, q)$$

for the same working state q. Therefore no state renaming transplants that exact U208 mixed-entry motif. This does not exclude more general contextual sharing or a different source allocation.

2.4 The arithmetic endpoint must be stated accurately

The upstream theorem inventory exports the internal statement

$$\mathsf{PA} \vdash \neg\,\mathrm{Halt}(M_{196}) \longrightarrow \mathrm{Con}(\mathsf{PA}),$$

and external Lean-metatheoretic nonhalting and independence results. It does not export the full internal biconditional. This is a statement about the supplied declarations, not a proof that the missing converse is false or unprovable. A uniform syntactic extraction of a PA refutation from a halted literal trace can establish the converse; external HF-model soundness alone does not supply that PA-internal theorem. The incoming packets contain additional written proof-translation arguments with explicitly stated implementation boundaries. None of the state-count experiments below fills that boundary by itself.

3 New combinations of previously separate reductions

The first new experiment composes previously supplied source changes, then reconstructs and checks their physical implementations. The changes are: the distributed-separation emitter; the diagonal-vacuity emitter; the shorter negation block; and the MP-success branch that returns its consequent in register H. These are combined as actual source trees, not by subtracting their separate state-count estimates.

The negation replacement changes a 47-instruction block `push; incT; cons` to a 20-instruction block `drainTtoX; incT; pair4`. Its mathematical equivalence requires the inherited scratch-zero entry invariant. The MP change removes 23 primitive instructions. These logical/source obligations are separate from the checks that a tape implements the resulting source program.

Emitter and changes	Sites	Raw states	Final states	Symbolic cases
Original	3812	202	196	4959
Original + MP	3789	202	196	4929
Original + short negation	3596	202	197	4671
Original + short negation + MP	3573	202	197	4641
Distributed	3773	201	198	4908
Distributed + MP	3750	201	198	4878
Distributed + short negation	3611	202	199	4692
Distributed + short negation + MP	3588	202	199	4662
Diagonal	3700	203	199	4815
Diagonal + MP	3677	203	199	4785
Diagonal + short negation	3484	203	200	4527
Diagonal + short negation + MP	3461	203	200	4497

Here "raw" refers to the straightforward allocation before contextual reader folds. For each source, 200 deterministic seeded fold orders were examined. The selected complete table passes every symbolic primitive contract, preserves the 42 explicitly checked engine rows needed by the register-block interface, and reaches the exact clean blank-start frame after 58,154 transitions. These are executable symbolic checks, not finite sampling of register magnitudes. They are also not a new proof-assistant kernel certificate or an exported PA refutation compiler.

The original and original-plus-MP rows retain 196 states; none is a strict improvement. The table gives a concrete reason the separate savings cannot simply be added. Shortening negation removes source work but disrupts useful contextual sharing, costing a physical state in these allocations. The reproducible subset is in `combinations/`; its replay writes fresh evidence into a separate `replayed/` directory.

3.1 Combining distribution with diagonal vacuity

A further source experiment combines the two logical emitter changes with one another. Write I for implication, U_b for universal quantification, and $N(X) = I(X, 1)$. For parameter codes a, b, c, set

$$\begin{aligned} D &= I(c, U_c(c)), \\ V &= I(M(a,b), U_c(M(a,b))), \\ P &= I(c, M(c,b)), \\ L &= I(M(a,b), P), \\ R &= I(M(c,b), c). \end{aligned}$$

The candidate gate is

$$N\big(I\big(D, I\big(V, U_c\big(I(U_b(L), I(U_b(R), N(P)))\big)\big)\big)\big)\,.$$

It combines the earlier diagonal clause with distribution of the quantifier over the separation conjunction, using the same original freshness guard. The preceding head order and one antecedent exchange are the previously checked distributed-emitter choices.

Exact macro expansion gives 189 tokens, compared with 201 in PA196, and 3661 source sites. Adding the MP rewrite gives 3638. Applying all six applicable shorter-negation replacements

gives 3499 sites, or 3476 with MP as well. These are exact source-size calculations; they do not establish a 189-state or a sub-196-state machine. A physical layout must retain the decoder-entry predicates, maximum-zero-run constraints, startup, and full symbolic runtime contracts. The source combination also inherits the outstanding uniform proof-translation and internal endpoint obligations of its constituent logical changes.

The joint word admits a concrete variable-arity dictionary whose ordinary binary concatenation cost is 66 after accounting for an already available two-increment block. A 60-second mixed-integer search supplied this incumbent; it did not prove optimality. The full witness is retained in `combinations/joint_dictionary/grammar66.json`. It is not an accepted physical layout. In that fixed dictionary, the first-part chain through production 208 forces a run of at least seven consecutive zero path bits: three come from the existing copy-A block, and four from mandatory left-child descents. The inherited reader bound is six. The independent `check_obstruction.py` verifies the exact 189-token word, cost 66, use of the offending production, and the seven-zero lower bound without rerunning a solver or trusting a random association. Thus this particular dictionary cannot be repaired merely by selecting a different association of its fixed ordered productions. Changing the productions or the reader remains possible. This local obstruction is not a lower bound for all 189-token emitter layouts, much less for all 195-state consistency machines.

4 The calculus relevant to the new logical experiments

For clarity, write $p(a,b) = \binom{a+b+1}{2} + b$. PA196 uses

$$I(a,b) = 4p(a,b), \quad M(a,b) = 4p(a,b)+1, \quad U(a,B) = 4p(B,a)+2, \quad E(a,b) = 4p(a,b)+3.$$

Code 0 denotes truth and code $1 = M(0,0)$ is the selected contradiction. The quantifier constructor's argument order is essential. Write $N(B) = I(B,1)$, $B \wedge C = N(I(B,N(C)))$, and $\exists x_a B = N(U(a,N(B)))$. An arrow below means I.

Apart from the restricted implicational head ρ_8, the six supplied heads are the following families, with parameters made unrestricted by the command compiler:

$$\begin{aligned}
&(\exists x_D\, U(D,H)) \to H,\\
&U(D,U(A,H)) \to U(A,U(D,H)),\\
&M(H,D) \to (E(H,A) \to M(A,D)),\\
&E(H,D) \to (M(A,H) \to M(A,D)),\\
&\exists x_D\, E(H,D),\\
&U(D,A) \to (U(D,A \to H) \to U(D,H)).
\end{aligned}$$

The restricted head is

$$((H \to d) \to D) \to ((D \to H) \to (A \to H)), \qquad d \le D.$$

The existing restricted-head derivation supplies classical implicational reasoning independently of the first quantifier head.

The gate, when $c \neq a,b$, is

$$F(a,b,c) = N\big(\Delta_c(a,b) \to (V(a,b,c) \to Q(a,b,c))\big),$$

where

$$\begin{aligned}\Delta_c(a,b) &= E(a,b) \to U(c, E(a,b)),\\ V(a,b,c) &= M(a,b) \to U(c, M(a,b)),\\ P &= c \to M(c,b),\\ S &= U\big(b, (M(a,b) \to P) \wedge (M(c,b) \to c)\big),\\ Q(a,b,c) &= U(c, S \to N(P)).\end{aligned}$$

When the guard fails it returns truth. Here the natural number c serves both as a formula code and as a variable label. The code-bounding lemma makes it fresh for the formula it codes. This interface is the reason a generic slogan about changing variable names is insufficient. The original construction and its formal proof inventory are given in [1, 2, 3].

5 Logical reductions, rejected deletions, and proof strength

This section separates three questions: equivalence of abstract proof calculi, the number of states obtained by compiling them, and formalization of the resulting arithmetic halting equivalence. A reduction in one of these measures does not settle the other two.

5.1 An equivalent split of the first quantifier head

Write $f = 1$ for the distinguished contradiction code and put

$$X = \mathsf{U}\big(D, \mathsf{U}(D, H) \to f\big).$$

The first PA196 head is

$$O(D,H) = (X \to f) \to H.$$

Consider replacing it by the two families

$$B(D,H) = (H \to f) \to X, \qquad E(A) = f \to A. \tag{Split}$$

All remaining heads, the guarded gate, modus ponens, and closed generalization are unchanged.

Split-head theorem. The original calculus and the calculus with (Split) derive exactly the same numeric formulas. The conversions are uniform finite implicational proof templates and hence induce primitive-recursive transformations of coded derivations.

Proof. The restricted ρ_8 family supplies K, S, and Peirce's law without using the first quantifier head or explosion. This is explicit in `M6/Logic/RestrictedImplication.lean`: its abstract `Rules` structure contains only modus ponens, the restricted row, and the three code-order conditions. The PA196 interpretation of those rules uses only its ρ_8 constructor. In particular, ordinary implicational deduction and

$$((P \to Q) \to P) \to P$$

are available before either side of the proposed conversion.

In the new calculus assume $p : X \to f$. Under the further assumption $n : H \to f$, the new head $B(D,H)$ gives X, then p gives f, and $E(H)$ gives H. Discharge n to obtain $(H \to f) \to H$ and apply Peirce's law with $(P,Q) = (H,f)$. Discharging p now gives $O(D,H)$.

Conversely, the old first head supplies explosion. For any A, take

$$Y = \mathsf{U}\big(0, \mathsf{U}(0, A) \to f\big), \qquad (Y \to f) \to A.$$

From the assumption f, K gives $Y \to f$; the displayed head gives A. Discharge f to obtain $E(A)$. To obtain $B(D, H)$, assume $n : H \to f$ and, inside this context, $q : X \to f$. The old head gives H, then n gives f, and the already derived $E(X)$ gives X. Discharge q, apply Peirce's law with $(P, Q) = (X, f)$, and discharge n. The result is $(H \to f) \to X$, as required. No quantifier rule occurs in either conversion. □

The accompanying `check_split_head.py` constructs exact Hilbert proof DAGs and checks every node using K, S, Peirce, assumptions, and modus ponens. The new-to-old and old-to-new templates each have 77 nodes; the old-to-explosion template has 17 nodes. These counts measure the particular checked certificates, not minimal proof lengths. Their status is a finite syntactic verification over the displayed derived implicational basis, not a full new Lean build of the tape machine.

Keeping the explicit explosion family is essential. The contraposed head $B(D, H)$ alone is not a replacement justified by classical contraposition: that argument already assumes that code 1 acts as falsity. Indeed, interpret implication materially, every membership atom as true, every equality atom as false, and every quantifier as the identity operation on truth values. Code 0 is true, code 1 is true, and code 3 is false. Every retained head and the guarded gate, together with $B(D, H)$, is valid under this evaluation; generalization preserves validity. But $1 \to 3$ is false. The proof above identifies the exact additional family that repairs this obstruction.

5.1.1 The corresponding command words

The split adds one head. An axiom command consequently pushes ten items: the saved top, one generalization, ρ_8, and seven heads. The guarded gate remains the new top. If S denotes Shift, the corrected generic stack words are

$$\begin{aligned} \text{pop}: &\quad \mathsf{AX}(0,0,0)\, S^{11}, \\ \text{push}: &\quad \mathsf{AX}(0,0,0)\, S^{9}\, \mathsf{MP}(0), \\ \text{discard}: &\quad \mathsf{MP}(1)\, \mathsf{AX}(1,0,0)\, S^{10}\, \mathsf{MP}(0). \end{aligned}$$

The generalization snapshot identity becomes

$$\mathsf{AX}(c,x,y)\, S^{10}(t,w,d,h) = \big(\mathsf{U}(d,h), w, 0, t\big).$$

For a head numbered j from the top among the eight nongeneralization heads, $1 \le j \le 8$, the generic selector is: discard, execute AX with the desired arguments, apply S^{j+1}, and pop $9 - j$ times. The gate selector discards, executes AX, and pops ten times. These are the PA196 stack identities with the push count increased by one; the position of each formula must also follow the chosen emission order. The two new head families have implication as their outer constructor, so their codes are 0 modulo 4 and cannot equal the halt code 1. Taking a clean store $d = h = 0$ makes the parameters of $B(D, H)$ and $E(A)$ unrestricted.

These command-word adjustments are necessary even when the emitted formula families are logically equivalent. A candidate table that changes the output count but retains the old selectors has not thereby inherited the original certificate compiler.

5.2 Why neither membership rewrite can simply be deleted

The two direct rewrite families are

$$L(a,b,c): \quad \mathsf{E}(a,b) \to (\mathsf{M}(a,c) \to \mathsf{M}(b,c)),$$
$$R(a,b,c): \quad \mathsf{E}(a,b) \to (\mathsf{M}(c,a) \to \mathsf{M}(c,b)).$$

They correspond to PA196 heads 3 and 4, with the harmless antecedent exchange used in head 3. The following countermodels rule out the proposal to delete either one and derive it from all the remaining numeric axiom families.

Let $\mathrm{Bit}(s,k)$ mean that bit k of $s \in \mathbb{N}$ is one. Let f_1 exchange 0 and 1, and let every other f_i be the identity. Thus every f_i is a finite-support permutation. Variables are assigned natural-number values x_i; implication and quantifiers have their ordinary classical meanings, and $\mathsf{E}(a,b)$ means $x_a = x_b$. Numeric code 0 denotes truth, as in the usual decoder.

Delete L. Interpret membership by

$$\mathsf{M}(a,b) \quad \Longleftrightarrow \quad \mathrm{Bit}(f_a(x_a), x_b). \tag{L-model}$$

The family R is valid because equal values in its second membership coordinate can be substituted. The quantifier heads, existential equality witness, ρ_8, and both atomic vacuity families are valid. Moreover, code 1 still denotes $\mathrm{Bit}(x_0, x_0)$, which is always false.

The guarded gate remains valid for every formula instance, including the case where its two ordinary labels coincide. When $a \neq b$, the values of the bound variable satisfying $\mathsf{M}(a,b)$ form the finite bitset of $f_a(x_a)$. When $a = b$, they form

$$\{n : \mathrm{Bit}(f_a(n), n)\}.$$

This set is finite: outside the finite support of f_a, the condition is $\mathrm{Bit}(n,n)$ and is false. Add the distinguished point x_b to this finite candidate set and retain precisely the elements satisfying the gate's arbitrary formula. If s codes the selected finite set, assign $x_c = f_c^{-1}(s)$. The guard $c \neq a, b$ and the self-freshness of numeric formula code c ensure that this assignment changes neither the old parameters nor the selected property. It therefore supplies exactly the required pointed separator. Equality and membership vacuity under the same guard are immediate.

Nevertheless, assign $x_1 = x_2 = x_3 = 0$. Then $\mathsf{E}(1,2)$ and $\mathsf{M}(1,3)$ hold while $\mathsf{M}(2,3)$ fails. Thus $L(1,2,3)$ is false and cannot be derived from the retained families.

Delete R. Instead interpret membership by

$$\mathsf{M}(a,b) \quad \Longleftrightarrow \quad \mathrm{Bit}(x_a, f_b(x_b)). \tag{R-model}$$

The family L is now valid. All other retained heads, both atomic vacuities, and the false interpretation of code 1 remain as above. For $a \neq b$, the gate's candidate domain is the inverse image under f_b of the finite bitset of x_a. For $a = b$, it is the finite set

$$\{n : \mathrm{Bit}(n, f_b(n))\}.$$

After adding the distinguished point and filtering by the arbitrary property, assign to x_c the bitset of the images of the selected values under f_b. This proves the entire gate schema in this model.

Set $x_1 = x_2 = 0$ and $x_3 = 2$. Then $\mathsf{E}(1,2)$ and $\mathsf{M}(3,1)$ hold, since bit 1 of 2 is one, while $\mathsf{M}(3,2)$ fails, since bit 0 of 2 is zero. Hence $R(1,2,3)$ is not derivable from its retained families.

These are generalized semantics for the actual numeric calculus, not ordinary structures with one uniform binary predicate. That distinction is deliberate: the calculus's primitive proof rules are modus ponens and closed generalization. An unrestricted substitution rule for arbitrary numeric formula codes is not available. Since every retained axiom family is valid and the two primitive rules preserve validity, these semantics suffice to prove nonderivability. They exclude a *derived-head justification* for either deletion. They do not exclude an entirely new interpretation proving that a smaller calculus has the same consistency strength.

The script `check_rewrite_countermodels.py` verifies the two explicit countervaluations and finite controls of the constructions. For labels $0, \ldots, 4$ and parameter values $0, \ldots, 7$, it checks 8,000 retained-rewrite cases and 23,056 separator cases for each model, exhausting every property restriction on each relevant finite candidate set. This is a check of the stated finite ranges. The full-schema argument is the mathematical finite-support construction above.

5.3 What the iPRA work can contribute

The native scalar core in the supplied `ConciseIPRA` project [6] is bi-interpretable with intuitionistic primitive recursive arithmetic. Using that core unchanged would target a different consistency statement. Its most direct useful modification is precise: retain its native terms, formula constructors, logical rules, and freshness conditions, but extend its progress rule from literal equations to arbitrary native formulas:

$$\frac{\Gamma, B(z) \vdash B(Sz)}{\Gamma \vdash Q_y B(y)}, \qquad Q_y B(y) \equiv B(0) \to \forall y\, B(y), \qquad z \text{ fresh.} \tag{Full progress}$$

The ordinary realization of this extended core is sound in Heyting arithmetic, using full formula induction. Conversely, the existing native definitions of the logical connectives and primitive-recursive functions, together with (Full progress) followed by closure elimination, supply induction for every translated Heyting-arithmetic formula. This gives a natural HA-strength source calculus. The usual negative translation from PA to HA and the inclusion of intuitionistic proofs into classical proofs then supply the intended consistency comparison, once implemented on the selected proof codes.

This is a concrete high-level avenue, not a delivered smaller tape machine. A compilation must still count the implementation of terms, substitution, freshness, contexts, certificate enumeration, and blank-tape startup. Python token counts and numbers of logical rule families do not determine binary single-tape state counts.

5.4 The reverse internal PA implication

Let H_{196} be the exact arithmetic halting sentence for the literal PA196 table, and let $N_{196} = \neg H_{196}$. The upstream declaration `PA196.Arithmetic.pa_proves_nonhalting_consistency` proves

$$\mathrm{PA} \vdash N_{196} \to \mathrm{Con}(\mathrm{PA}).$$

The source's published theorem inventory does not include the converse. Its external nonhalting proof uses a hereditarily finite-list model in Lean's metatheory. That external soundness proof is not itself a PA proof of the reverse implication.

There is a direct syntactic route to the missing direction. Translate numeric membership to $\mathrm{Bit}(x_a, x_b)$, numeric equality to arithmetic equality, and decode the transposed quantifier constructor correctly. For every arithmetic formula $\varphi(x)$, PA proves

$$\forall a, b\, \exists s\, \forall x \quad \mathrm{Bit}(s, x) \leftrightarrow \big((\mathrm{Bit}(a, x) \vee x = b) \wedge \varphi(x)\big). \tag{PC}$$

For a detailed template, set $m = \max(a, b+1)$ and use full induction on $k \le m$ to construct a binary sum agreeing with the desired property on the first k bits. At stage k, excluded middle for $\varphi(k)$ determines whether to add 2^k. The implication $\mathrm{Bit}(a, x) \Rightarrow x < a$ bounds every possible selected value. The formula φ is inserted syntactically into this one induction proof template. Consequently the map from φ to its PA proof of (PC) is primitive recursive; it does not evaluate φ or introduce a uniform arithmetic truth predicate.

Pointed comprehension supplies PA proof templates for the guarded gate. The other heads have fixed logical proof templates. A primitive-recursive induction on a numeric derivation then translates any derivation of code 1 into a PA refutation. The command interpreter can carry these proof codes in its register and stack invariant. At a halt test $t + h = 1$, one of the two proved codes is 1, so a refutation is extracted. This replaces semantic truth by syntactic proof objects throughout the invariant.

The physical step is an internal version of the existing external `blank_halting_iff_source` theorem. Positive forward macro simulation is sufficient: if the literal machine halts by time n and the source does not halt in its first $n+1$ steps, concatenate $n+1$ positive macro simulations to obtain a literal execution longer than n, contradicting deterministic halting. This uses bounded induction and the macro contracts, not an unbounded termination argument. A completed implementation of these constructions would prove

$$\mathrm{PA} \vdash H_{196} \to \mathrm{Pr}_{\mathrm{PA}}(\ulcorner \bot \urcorner), \quad \text{hence} \quad \mathrm{PA} \vdash \mathrm{Con}(\mathrm{PA}) \to N_{196}.$$

This paragraph describes the proof construction and identifies its formalization tasks. It is not a claim that the missing arithmetic Lean theorem has been supplied or checked in this work.

5.5 A stronger obstruction to replacing the first head by reflexivity

Even adding explicit explosion does not justify replacing the old first head by the shorter family

$$\mathsf{U}(D, H) \to H. \tag{T}$$

Here is a countermodel to that proposed derived-head argument. A world is a natural-number assignment together with a stage $s \in \{0, 1\}$. The quantifier $\mathsf{U}(i, \varphi)$ ranges over all worlds obtained by changing x_i arbitrarily and replacing s by any $s' \ge s$. Implication is material at each world. Equality is ordinary numerical equality. At stage 0 every membership atom is false; at stage 1, $\mathsf{M}(a, b)$ has its usual bit interpretation $\mathrm{Bit}(x_a, x_b)$. Code 1 is false at both stages.

The accessibility relation for each quantifier is reflexive, so (T) holds. Relations for different labels commute, since their two coordinate changes and their final nondecreasing stage can be chosen in either order. Distribution holds for universal modalities. The equality witness holds by choosing the quantified value equal to the other value. Both membership rewrite families hold: at stage 0 their membership antecedents are false, and at stage 1 they are ordinary equality substitutions. Atomic vacuity at a distinct label also holds: equality is unchanged; a true membership atom already lies at stage 1 and remains unchanged. Every explosion instance holds because code 1 is false.

The guarded gate is valid at both stages. Its equality and membership vacuities were just checked. To refute its universal Q clause, choose the separating variable at stage 1. All its subsequent quantifiers then remain at stage 1, where ordinary finite pointed separation supplies a witness. The formula-code freshness property has its usual meaning at that stage. The separating set makes both S and P true, so Q is false and the whole gate is true.

But take a stage-0 world with $x_2 = 1$, $x_3 = 0$, put $H = \mathsf{M}(2, 3)$, and take $D = 4$. Here H is false. After advancing to stage 1, $\mathsf{U}(4, H)$ is true, since changing x_4 leaves the true assertion

$\mathrm{Bit}(1,0)$ unchanged. Hence $\exists x_4\, \mathsf{U}(4,H)$ is true at the initial world, and the old first head is false there. Modus ponens and closed generalization preserve validity in this semantics. Consequently the first head is not derivable from (T), explosion, and the remaining numeric axiom families. This is a stronger obstruction than merely observing that a proposed replacement fails to make code 1 behave as falsity.

6 New grammar screens and their limitations

The new split-head equivalence was tested as an actual changed emitter. The macro word has 205 items instead of 201: the contraposed modal head has the same constructor count as the old first head, while the separate explosion family adds a four-item expression. Seven heads can be emitted in different orders, and their parameter roles can be renamed. These are sound source-level choices when the changed command selectors are used.

A second tested presentation uses the positive modal form

$$H \to U\big(D, U(D, H \to 1) \to 1\big)$$

together with explosion. Substituting $H \to 1$ into the contraposed form and using $H \to ((H \to 1) \to 1)$ derives this positive form. Conversely, substitute $H \to 1$ into the positive form and use double-negation elimination, available with explosion and classical implication, under the retained quantifier-monotonicity rule. Thus it is another safe presentation with the same number of constructors but a different emission order.

The third family pads the formula parameter in the first head with formulas equivalent to it under the unchanged implicational basis. Examples are $0 \to H$, $(H \to H) \to H$, and $(H \to 0) \to H$. Code 0 is already derivable truth, so these equivalences do not use the first quantifier head. Quantifier monotonicity transports them through its two quantifiers. Longer words can occasionally share more subtrees, so this is a legitimate state-optimization question despite the additional symbols.

Screen	Seeded trials	Distinct words	Smallest raw AX DAG
Contraposed modal head plus explosion	50,000	22,057	74
Positive modal head plus explosion	75,000	26,433	74
Truth padding of the original head	50,000	24,882	68

The search uses the inherited repeated-subword dictionary and a randomized pair-replacement heuristic with fixed seed 2026111342. Before each run, the original compiler is checked to produce 196 states from the original 68-node body. The two split families never reached the heuristic's 72-node physical-allocation cutoff. Among the padding samples, 499 candidate parses reached that cutoff, but every one failed the imposed maximum-six-zero-run or 46-bit-path screen. The padding family includes the unchanged parameter, so its score of 68 is not claimed as a new grammar improvement.

Consequently no layout in these three screens was accepted for physical pricing. The recorded `best_physical` field is `null`; it is not a numerical state bound. The report deliberately does not convert 74 raw grammar nodes into a purported minimum machine count. The searches are finite heuristics, do not exhaust associations or logical presentations, and do not rule out a better implementation of the proved split-head calculus.

The portable script is `grammar/split_head_search.py`; its minimal support sources and licenses are included. The exact receipts are under `results/`. The two directions of the split-head proof are also serialized in `logic/split_head_certificates.json`. A separate checker reads those

stored DAGs without importing their generator and validates every axiom instance, assumption, prior-node reference, modus ponens step, and final conclusion.

7 Additional searches using source-derived invariants

No state reduction was obtained in the searches described in this section. Their value is to examine several transformations that are not covered by a literal action-preserving quotient. The supplied programs, frozen inputs, positive controls, adverse controls, and receipts are sufficient to rerun the experiments without a network connection.

7.1 A stronger finite context invariant

A state merger need not work on every possible local tape word. It suffices that its simulation contract holds on every local word that can occur in the original blank computation. A bounded-window closure is one sound approximation: after a head movement, the bit newly exposed at the edge of the window is allowed to be either symbol. This deliberately forgets correlations between the path reader and the register program.

We obtain a stronger approximation from the complete source-level coverage. The inherited symbolic interpreter checks all $4,959$ zero/positive cases for the $3,812$ source sites, with arbitrary natural register values and gap. Every ordinary decoder transition exposes a finite symbolic tape prefix. We enumerate the local words represented by that prefix, allowing all lengths permitted by a variable unary run and all bit patterns at an opaque register box. Correlations between variable lengths may be forgotten; this enlarges the set and is safe. States used by an opaque engine contract, and states whose self-loops may be skipped by the interpreter, retain their full coarse blank-window contexts. Finally, all $58,154$ startup transitions are included, and the exact final clean frame is checked.

Radius	Source-derived contexts	Coarse blank contexts
2	2,412	2,848
4	17,343	25,827
6	161,614	270,067

The mathematical coverage argument is induction over the startup and the source-instruction contracts. It is not a new kernel-checked invariant in this package. In particular, the imported Python interpreter uses the established register-engine contracts as an oracle. The construction does not infer a universal property from sampled register values.

7.2 Moving the alignment boundary

A conventional local quotient identifies a deleted control a with a retained control b, then demands that a positive child block match one parent step from either control. We also tried deleting a at its predecessor boundary. If a has one incoming row, that row may be replaced by an arbitrary write/direction/target triple. The child is compared with the old two-step block across a, or with a later old endpoint. This allows a different incoming write or head position without treating control a itself as a common boundary.

The unique-predecessor search considered 91 states and $71,344$ replacement actions. A further search considered all 195 noninitial working states, trying incoming rows separately and stopping a deletion when one predecessor had no bounded witness. With radius-six coarse contexts and a

64-step bound, it tested 157, 584 individual actions and found no complete deletion. This is not an exhaustive search of mutually dependent simultaneous changes to several predecessor rows.

7.3 Storing control information in the head or tape

Two further templates change the representation of an intermediate configuration. Write a configuration as (q, h, T) and let $\pi(a) = b$, with π otherwise the identity. We exclude deleting the initial state. Here T is the set of cells containing 1, and $[P]$ is 1 when P holds and 0 otherwise. Suppose all incoming transitions to a from other states move in direction $d \in \{-1, 1\}$.

For a head-offset encoding use

$$\Phi(q, h, T) = \big(\pi(q),\ h - 2d\,[q = a],\ T\big).$$

Every incoming direction to a is reversed. Its new head position is then exactly the one specified by Φ. One row of b is synthesized, and the finite checker requires a positive child block from $\Phi(C)$ to $\Phi(C')$ for each permitted parent transition $C \to C'$.

For a neighboring-bit encoding use

$$\Psi(q, h, T) = \begin{cases} (b, h, T \triangle \{h - d\}), & q = a, \\ (q, h, T), & q \neq a. \end{cases}$$

Here the symmetric difference complements that tape cell. Incoming rows to a complement their written bit, and one retained row is synthesized. The child block must restore or correctly transport the tape difference. These templates can carry a deleted state in physical information that an ordinary same-head, same-tape quotient does not preserve.

The controls include explicit three-step examples for both encodings. Changing the restoration bit, or the required helper direction, invalidates the corresponding witness. For PA196, the recorded head-offset and neighboring-bit searches each tested 52, 590, 720 variants and found none. The packaged full replay repeats both encodings at radius six with the same negative outcome; its combined receipt also records the positive and adverse controls.

7.4 Results and limitations

The expanded one-retained-row search at radius four tested 59, 928, 960 patches with a 64-step bound and found no survivor. At radius six, the two-retained-row search examined 29, 964, 480 first-row actions. Its unknown-second-row prefilter retained 81, 874 choices; 64, 189, 216 complete row pairs were checked, again without a survivor.

These results are restricted negative search outcomes. They do not establish minimality of the 196-state machine. A different axiom calculus, path layout, register encoding, initialization, wider window, longer matching block, or more extensive simultaneous rewrite remains outside these searches. No 195-state table is asserted by this package, and none of the receipts by itself proves a PA-internal consistency equivalence.

8 How to reproduce and interpret the archive

Start with `README.md`. The default command

```
python3 verify.py
```

checks the artifact manifest, the complete selected-table comparison, the new finite Hilbert certificates and countermodel controls, and the recorded evidence summaries. More expensive source and physical searches have separate commands, so reproducing the report does not silently launch a large search. Input snapshots and source identities are retained alongside the code. The comparison implementation is inherited from the incoming research and attributed accordingly; the replay binds its inputs to the current repository snapshot.

The unchanged upstream source/table check was also run successfully: `scripts/check.sh` checked all 392 PA196 rows, the Palomar source checks, and 41 generated Lean files. Its output is retained in the archive. This is the default source check, not the optional full Lean build and axiom audit. The complete upstream Foundation/Mathlib proof development was not rebuilt for this report.

The literal reference table is included for reproducibility. Every experimental table is labeled with its actual count and status. A passing finite startup, a smaller source program, an optimistic grammar score, or a source-level proof does not by itself certify a smaller PA-consistency machine. No file in the archive is labeled as an accepted 195-state construction.

9 Further research questions

Can the tape marker and source layout be optimized together? The return scanner is a concrete place where one state would suffice. Shortening its zero-run counter changes how temporary path flags are recognized, not just how clean source paths are recognized. A successful redesign must constrain intermediate paths or change the flag encoding and re-establish the engine contracts. The failed direct deletion is useful because it identifies that interface precisely.

Can an offset simulation enable a deletion? Most established folds return to the same head position. A simulation relation that permits a fixed state-dependent head displacement enlarges the design space. It must account for writes on both sides, all incoming paths, startup, and positive progress. The bounded offset screens in this investigation do not exhaust such relations.

Which logically equivalent axiom presentation minimizes the compiled decoder? The split-head theorem supplies exact finite proof translations, and the combined diagonal/distributed emitter shortens the macro word. Neither has yet lowered the validated physical count. The promising objective is the size of a verified decoder allocation, including available history tests and marker constraints. Macro-token count alone is an inadequate objective.

Can the missing internal implication be completed on the existing exact trace predicate? The proof-producing bit-comprehension argument and bounded positive simulation give a concrete route. Implementing that route once for PA196 would make the precise calibration reusable for descendants whose source transformations are formally certified. This would strengthen the foundation of the optimization work even without a state saving.

Does the native HA-strength core compile competitively? The proposed extension of the ConciseIPRA progress rule changes the induction strength in the required direction. The next useful artifact would be an explicit low-level interpreter and state budget for that core, including certificate representation and enumeration. Its compact abstract proof checker alone cannot answer the state-count question.

References

[1] James Chang. *Nonhalting and PA-independence for a 196-state Turing machine*. Paper and source at revision `407cdc356f6b586635e71ac92be394255f4bfea4`. `https://github.com/jamesyc/BusyBeaver/tree/407cdc356f6b586635e71ac92be394255f4bfea4`.

[2] James Chang. *Formal theorem statements*, `docs/result.md`, same PA196 revision. `https://github.com/jamesyc/BusyBeaver/blob/407cdc356f6b586635e71ac92be394255f4bfea4/docs/result.md`.

[3] James Chang. The requested PA209 snapshot, `69d98241a721ead20b4f0b17df9062950b40fbfe`. `https://github.com/jamesyc/BusyBeaver/tree/69d98241a721ead20b4f0b17df9062950b40fbfe`.

[4] Vladimir Reshetnikov. ProveIt, incoming research archives, revision `dc523e7b06ffb414093c90aa035500ca283e5f75`. `https://github.com/VladimirReshetnikov/ProveIt/tree/dc523e7b06ffb414093c90aa035500ca283e5f75/docs/incoming`.

[5] ProveIt. Imported PA196/PA209 comparison and reduction-search dossier, same revision, including `Computability/BusyBeaver/Research/PA/pa196-vs-u208/COMPARISON.md`. `https://github.com/VladimirReshetnikov/ProveIt/tree/dc523e7b06ffb414093c90aa035500ca283e5f75/Computability/BusyBeaver/Research/PA/pa196-vs-u208`.

[6] Vladimir Reshetnikov. ProveIt, *ConciseIPRA*, same revision, including the native scalar rules and their interpretation. `https://github.com/VladimirReshetnikov/ProveIt/tree/dc523e7b06ffb414093c90aa035500ca283e5f75/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA`.